



RAMCO INSTITUTE OF TECHNOLOGY

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Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical and Electronics Engineering

Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. EEE

Course Code & Title: EE8601 Solid State Drives

Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

❖ **Unit / Topic:** Unit I / Typical load torque characteristics

- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 4
- **Activity Chosen:** Mind map

• **Justification:**

Mind mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mind mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches.

Time Allotted for the Activity: 05 minutes

• **Details of the Implementation:**

Mind mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mind map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

CO – PO / PSO mapping:

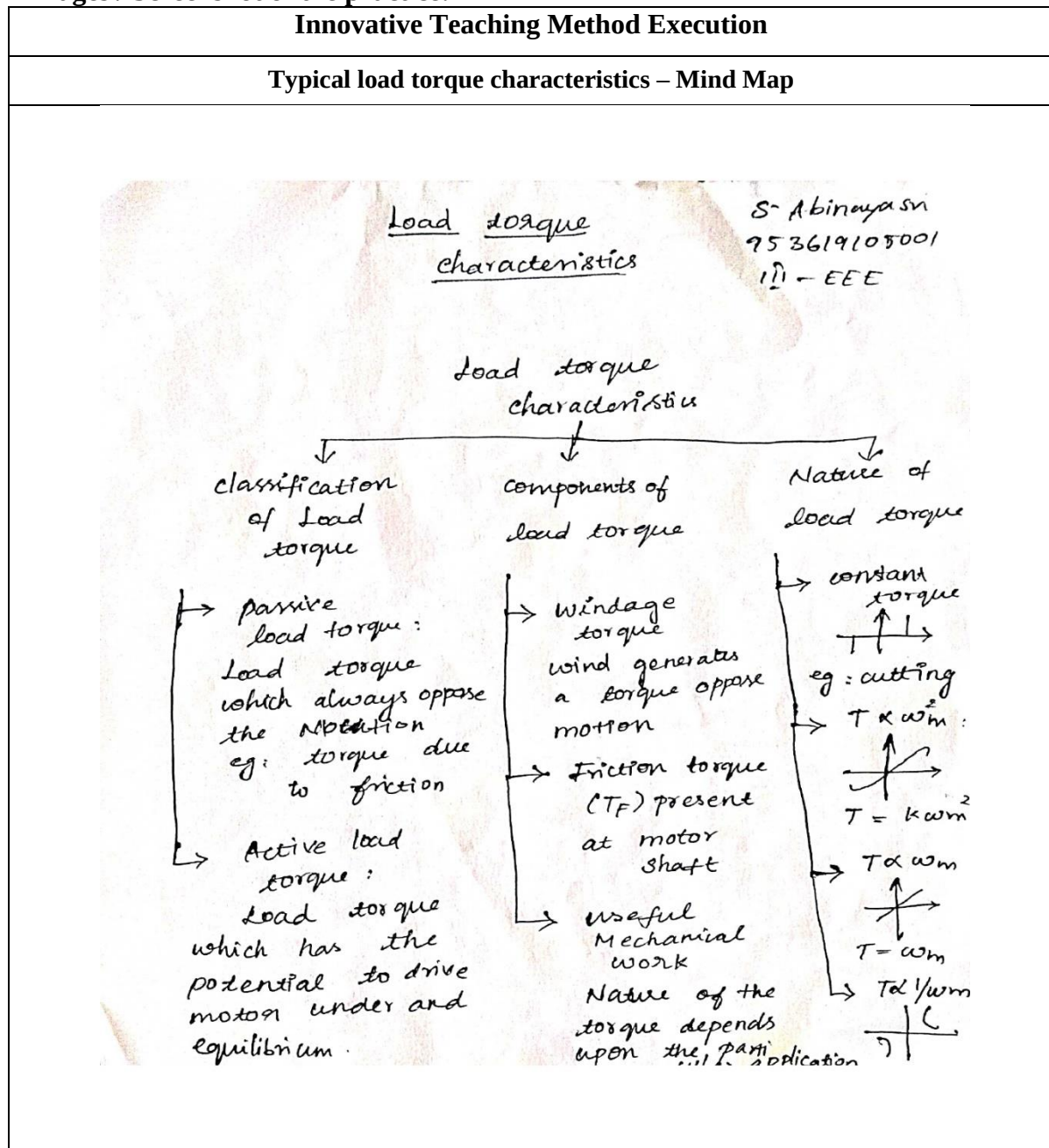
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	-	-	-	-	-	-	1	-	-	1	1	-	-

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- Images / Screenshot of the practice:



- Reflective Critique:

- ❖ Feedback of practice from students and other stakeholders:

Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ Benefit of the practice:

The benefits of Mind Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mind map made slow learners to remember the information more quickly.

Challenges faced in implementation:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to draw the map to Indicate relationships between the topics in typical load torque characteristics. I planned the activity for 05 minutes only. But in real scenario it takes 10 minutes to complete this activity.

References:

1. Gopal K.Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, 1992.
2. Bimal K.Bose. Modern Power Electronics and AC Drives, Pearson Education, 2002.
3. R.Krishnan, Electric Motor & Drives: Modeling, Analysis and Control, Pearson, 2001.

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Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

❖ Unit / Topic: Unit II / Applications

- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen:** One minute paper
- **Justification:**

Applications of dc motor drive in wide range of industries due to their constant speed operation, dynamic response and high starting current. Since, one minute paper activity provides a conceptual bridge between successive class periods. Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

Time Allotted for the Activity: 02 minutes

• **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

CO – PO / PSO mapping:

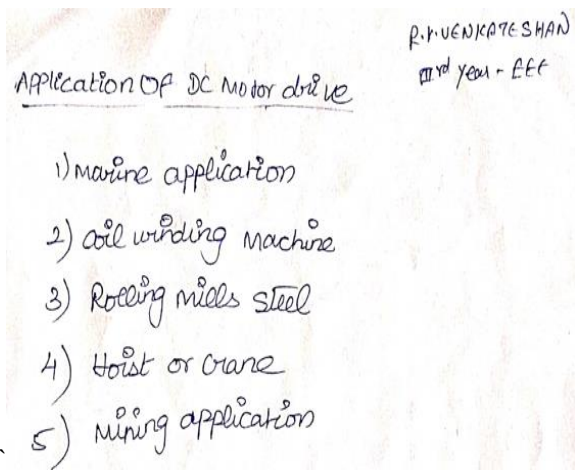
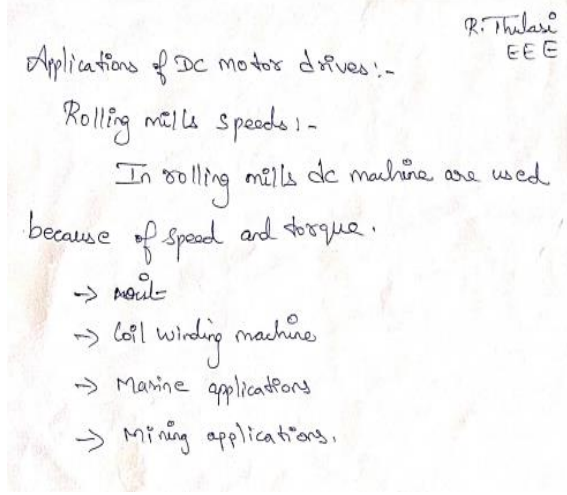
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	1	-	1	-	-	-	1	-	-	1	1	-	-

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

• **Images / Screenshot of the practice:**

Innovative Teaching Method Execution	
Applications of DC motor drive – One minute paper	
 Application of DC Motor drive 1) marine application 2) coil winding machine 3) Rolling mills steel 4) Hoist or crane 5) mining application R. VENKATESHAN IIIrd year - EEE	 Applications of DC motor drives:- Rolling mills speeds:- In rolling mills dc machine are used because of speed and torque. → hoist → coil winding machine → Marine applications → Mining applications. R. Thilasi EEE

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

❖ **Benefit of the practice:**

- ✓ Students can able to attend the question even in the questions are in indirect form.
- ✓ Students can able to explain the concepts in examination without any confusion.

❖ **Challenges faced in implementation:**

- ✓ Time utilization for conducting activity.

References:

1. Gopal K.Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, 1992.
2. Bimal K.Bose. Modern Power Electronics and AC Drives, Pearson Education, 2002.
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Course Code & Title: EE8601 Solid State Drives

Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

❖ **Unit / Topic:** Unit III / Rotor resistance control

• **Course Outcome:** CO 3

• **Topic Learning Outcome:** TLO 9

• **Activity Chosen:** Virtual Lab

• **Justification:**

Virtual lab refers to a virtual teaching and learning environment aimed at developing students' laboratory skills. As one of the most important eLearning tools, they allow the student to conduct various experiments without any constraints to place or time, in contrast to the constraints of real labs. By this student can visualize the effect of rotor resistance in the speed control of Induction motor.

Time Allotted for the Activity: 15 minutes

• **Details of the Implementation:**

The students were instructed to make the proper connection by clicking the node as instructed. If the wire is misplaced, click the node number to detach the nodes' wire. Switch on the MCB. After that, click anywhere on the autotransformer and slide the knob of the Rheostat to change the resistance. Take the readings from Rotor, Voltmeter, Ammeter and Watt-meter. Click "Add to table" button to insert the values in the table. Similarly, after taking minimum 6 readings, click the "Graph" button.

CO – PO / PSO mapping:

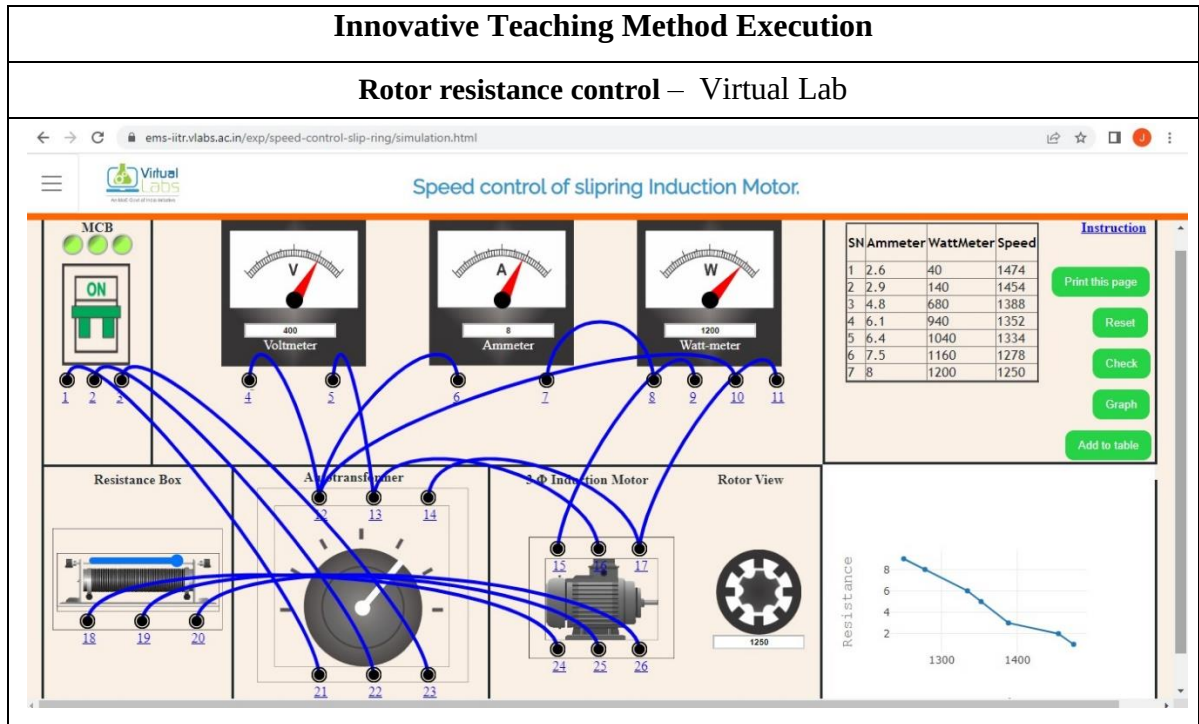
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	2	1	1	-	1	-	-	-	1	-	-	1	1	-	-

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO5
	1
Justification for correlation	The students can effectively use the virtual lab.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
 - ✓ Also they can able to visualise the effect of speed with respect to rotor resistance, therefore they easily understand the concept.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The process of simulating a scenario to practice different responses and actions to a real life situation is extremely effective in knowledge retention. This is because knowledge isn't in theory – the user needs to apply it in a real-life situation. Simulations can be slowed down to study behavior more closely. Conditions can be varied and outcomes investigated. Also Critical situations can be investigated without risk.

- ❖ **Challenges faced in implementation:**

I planned the activity for 15 minutes. But in Class room it takes 25 minutes.

References:

1. Gopal K.Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, 1992.
2. Bimal K.Bose. Modern Power Electronics and AC Drives, Pearson Education, 2002.
3. R.Krishnan, Electric Motor & Drives: Modeling, Analysis and Control, Pearson, 2001.

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Academic Year 2021 – 2022 (Even Semester)

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Course Code & Title: EE8601 Solid State Drives

Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

- **Unit / Topic:** Unit IV / Three phase current source fed synchronous motor
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 15
- **Activity Chosen:** Think pair share
- **Justification:**
 - ✓ It helps students to think individually about a topic or answer to a question.
 - ✓ It teaches students to share ideas with classmates and builds oral communication skills.
 - ✓ It helps focus attention and engage students in comprehending the reading material.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Think-Pair-Share innovative practice conducted for III year EEE students, after explained the concept of voltage source inverter and current source inverter fed synchronous motor. First, I asked the students to think about the difference between VSI and CSI fed induction motor for 2 minutes. Then I make them as a pair to discuss their neighbour's and asked the students to discuss about pros and cons of VSI and CSI fed synchronous motor for 3 minutes. Finally I asked the one of the team to explain the concept to whole class for further discussion. The students from group share their points and participated in the discussion for 10 minutes.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	2	1	1	-	-	-	-	-	1	-	-	1	1	-	-

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

• **Images / Screenshot of the practice:**

Innovative Teaching Method Execution															
Three phase current source fed synchronous motor – Think Pair Share															
953619105009 - M. Jeyapriyadharshini, 953619105018 - Muthu Ramya R <table border="1"> <thead> <tr> <th>VSI (Voltage Source Inverter)</th><th>CSI (Current Source Inverter)</th></tr> </thead> <tbody> <tr> <td>* In VSI input voltage is kept constant</td><td>* In CSI input current is kept constant</td></tr> <tr> <td>* VSI is fed from a Dc voltage source having small or negligible impedance.</td><td>* CSI is fed with adjustable current source from a Dc voltage source of high impedance.</td></tr> <tr> <td>* Slow response than CSI</td><td>* fast response than VSI</td></tr> <tr> <td>* diodes are connected in parallel</td><td>* diodes are connected in series.</td></tr> <tr> <td>* PWM technique is used</td><td>* PWM technique is not used.</td></tr> <tr> <td>* It requires feedback diodes.</td><td>* does not require any feedback diodes.</td></tr> </tbody> </table>		VSI (Voltage Source Inverter)	CSI (Current Source Inverter)	* In VSI input voltage is kept constant	* In CSI input current is kept constant	* VSI is fed from a Dc voltage source having small or negligible impedance.	* CSI is fed with adjustable current source from a Dc voltage source of high impedance.	* Slow response than CSI	* fast response than VSI	* diodes are connected in parallel	* diodes are connected in series.	* PWM technique is used	* PWM technique is not used.	* It requires feedback diodes.	* does not require any feedback diodes.
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* It requires feedback diodes.	* does not require any feedback diodes.														

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

❖ **Benefit of the practice:**

Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating.

❖ **Challenges faced in implementation:**

I planned the activity for 10 minutes. But in Class room it takes 15 minutes.

References:

1. Gopal K. Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, 1992.
2. Bimal K. Bose. Modern Power Electronics and AC Drives, Pearson Education, 2002.
3. R. Krishnan, Electric Motor & Drives: Modeling, Analysis and Control, Pearson, 2001.

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Course Code & Title: EE8601 Solid State Drives

Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

- **Unit / Topic:** Unit V / Design of controllers: Speed controller – PI controller
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 19
- **Activity Chosen:** Sage and Scribe
- **Justification:**
 - ✓ The chosen topic will have the concepts related to design of various controllers for motor drive. Hence if the students are doing co-operative learning, then it is easier for them to understand the concepts. Sage and Scribe is one such cooperative learning activity.
 - ✓ The expected outcome is like a review process of learned concepts / points, Sage and Scribe is suitable activity.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**
 - ✓ The two students sitting adjacent are made to do the activity together. One student (Sage) will explain the concept for a while and other student (Scribe) will answer for the questions.
 - ✓ For the next question, the students will change their role. Likewise it will be repeated for all the questions.
 - ✓ The students will share the same piece of paper for doing this activity.
 - ✓ I clearly explained them that, since it is a cooperative activity both is responsible for the outcome.
 - ✓ At the end of the lecture hour, the students are made to explain their answers to the class. I have given the correct answers for the questions with explanation.

CO – PO / PSO mapping:

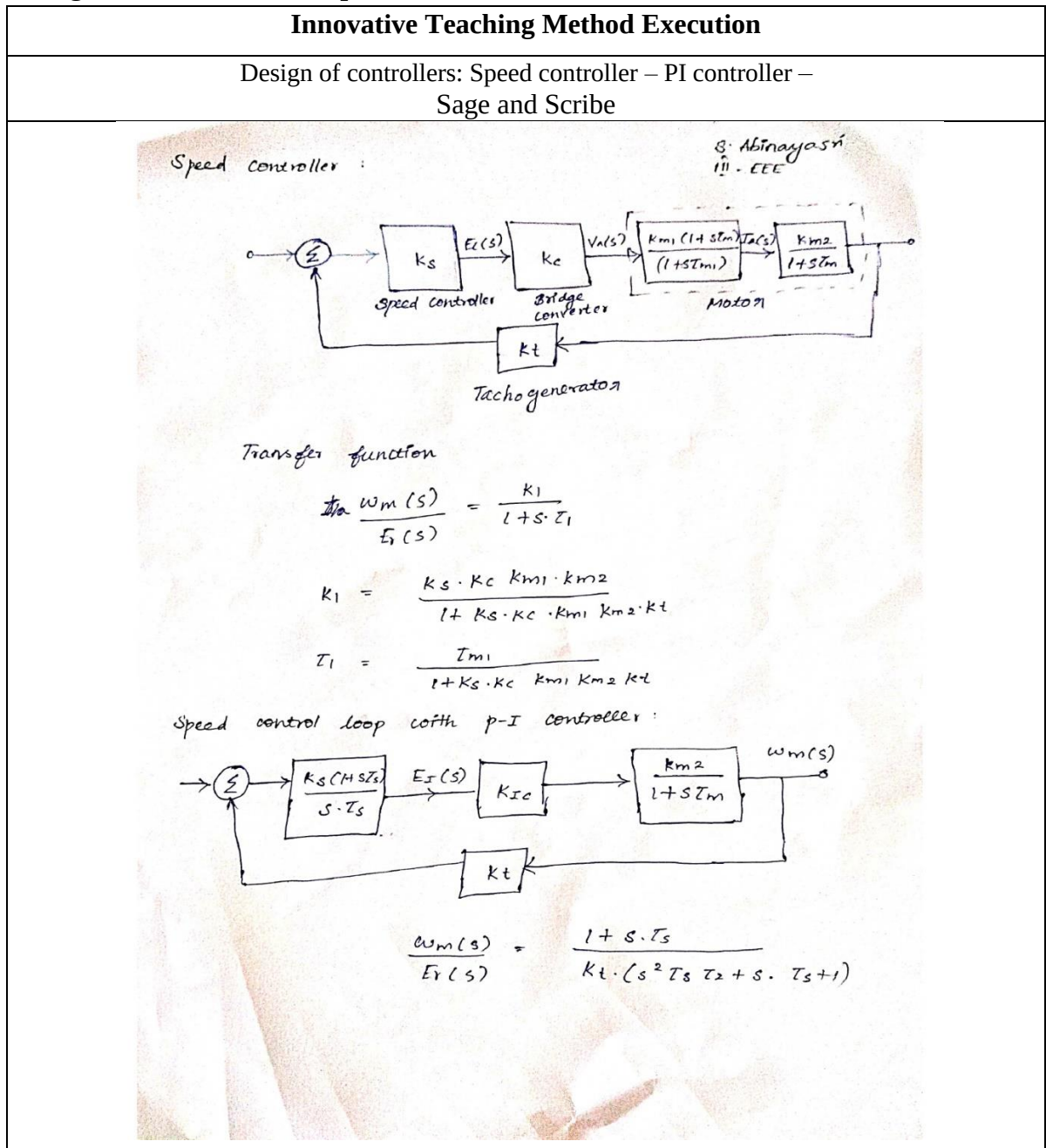
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CO5	2	1	1	-	1	-	-	-	1	-	-	1	1	-	-

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

• Images / Screenshot of the practice:



Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- ✓ The students have liked the session and they felt comfortable in doing this activity, since they are interacting with their neighbor. They did the work easily.
- ✓ They conveyed that more such activity will help them in understanding the concepts clearly and easily.

❖ **Benefit of the practice:**

- ✓ The students have understood the concepts clearly and it has been evident from the answers they have written on the answer paper.
- ✓ The students have presented the answers to the class. It cleared any doubt in understanding the important points in the chosen topic.

❖ **Challenges faced in implementation:**

- ✓ In few groups, both the students were slow learners; hence they find it very difficult to answer the questions. I helped them in answering the questions by providing them with simple hints.
- ✓ In the next time, while doing this activity, I should properly pair the students.

References:

1. Gopal K.Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, 1992.
2. Bimal K.Bose. Modern Power Electronics and AC Drives, Pearson Education, 2002.
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